ETSI Release 19 specification correction candidates

Generated by Codex model GPT-6-astra, 2026-10-04T19:56:35+02:00 (Europe/Paris).

Cross-checked by Claude Opus 5.5 (Anthropic, model ID claude-opus-5-5), 2026-10-04T20:12:00+02:00 (Europe/Paris).

NOTE — interoperability: these findings come from a desk review of the specification text only. None has been confirmed by interoperability testing between implementations, and none is an approved ETSI/3GPP change request. Where the text is ambiguous or self-contradictory, independent implementations may have chosen different readings (for example the protected-payload IV length in P04, or the MCVideo Video SSRC field ID in P27); confirm the behaviour with each peer before relying on a proposed correction.

The Source column gives the ETSI version (Release 19) of each specification reviewed. Page numbers are PDF pages of that version, counting the cover as page 1.

Protocol and XML schema corrections

ID	Source	Location	Current text or conflict	Proposed correction
P01	TS 24.282 V19.7.0	pp. 129, 438, 440–443, 628–631, 633; §§19A, D.8	MCData MBS text mixes `mcdata-mbs-usage-info+xml` and `mvdata-mbs-usage-info+xml`; service ICSI, schema root and namespace also use `mvdata`. §19A.3.2.2 points to MBMS Annex D.5.	Select consistent MCData MBS MIME, root and namespace identifiers across procedures, capabilities, schema and registration template; change the annex reference to D.8. The surrounding MCData text supports `mcdata`, but coordinate normative identifier changes.
P02	TS 24.281 V19.6.0, TS 24.282 V19.7.0, TS 24.379 V19.7.0	24.281 pp. 440, 590–591; 24.282 pp. 440, 628–629; 24.379 pp. 601, 735–736	UE join/leave procedures request one or more `<mbs-session-id>` elements without specifying the wrapper. All three schemas place status and ID inside one `<ue-session-join-notification>` and allow one ID.	Specify the wrapper in procedures and reconcile notification/ID cardinality with multi-session reporting.
P03	TS 24.281 V19.6.0, TS 24.282 V19.7.0, TS 24.379 V19.7.0	24.281 pp. 438, 441–442, 594–595; 24.282 pp. 438, 441–442, 631–632; 24.379 pp. 595–596, 598, 738–739	MBS procedures and semantics call `<mbs-session-infoType>` and `<eMBMS-bearer-infoType>` elements. Schemas define elements `<mbs-session-info>` and `<eMBMS-bearer-info>` with those types.	Use the element names in transmitted-element requirements and retain the `Type` names only for schema types.
P04	TS 33.180 V19.4.0	pp. 94–95, §8.5.4	The protected payload fixes IV length at 16 octets, but `DP_AES_256_GCM` specifies a 256-bit IV. Both cipher suites are defined as AEAD AES-GCM per RFC 5116 [43], whose nonce is 12 octets.	Align the encoded IV and both algorithm tables to 12 octets (96 bits), as required by the named AEAD_AES_128_GCM and AEAD_AES_256_GCM profiles in RFC 5116 §§5.1–5.2. Any other GCM nonce profile would need an explicit specification change.
P05	TS 33.180 V19.4.0	pp. 94, 174; §§8.5.4.1, F.1.5	Protected payload `DPPK-ID` is four octets; KDF input F.1.5 gives its length `L0` as `0x0017` (23 octets).	Set `L0` to the actual encoded DPPK-ID length, or define any different identifier representation explicitly.
P06	TS 24.379 V19.7.0	pp. 108, 510; §§6.3.2.1.8.1, 11.2.1.1.1	Imminent-peril authorisation is described as emergency-call authorisation. MONP reception says `port TBD as received 8809 messages`.	Say “imminent peril group call”; say `port 8809 as received MONP messages`, matching sending and §10.2.1.1.1.
P07	TS 24.581 V19.2.0	pp. 21–22, §§4.1.1.2, 4.1.1.4	A transmission participant is assigned control-server messages; a receiving participant is told to obtain priority from another user's Transmission Request, which it does not receive.	Correct the receiver-local stream-selection procedure: remove or rework the copied grant/notification actions, which belong to the control server. Specify how the receiver obtains the priority used to compare streams.
P08	TS 24.582 V19.0.0	pp. 15–17, §§6.1.1.2.2, 6.1.1.3.2	Reply generation requires a “Reply ID IE”; reception uses “InReplyTo Message ID” and 24.282 §15.2.11 “InReplyTo message ID.”	Name the generation field “InReplyTo message ID IE.”
P09	TS 24.545 V19.4.0	pp. 17, 74, 82; §§6.2.2.2.2, 7.3, 7.4.2	§7.3 mandates `report-id` in every `<report>`; the procedure includes it only for request-triggered reports and the schema marks it optional.	State the condition in §7.3 so event-triggered reports without a request ID are valid.

ID	Source	Location	Current text or conflict	Proposed correction
P10	TS 24.545 V19.4.0	pp. 74, 91, 95; §§7.3, 7.4.2, 7.5	`tRangeDirectionType` puts three `<xs:attribute ... minOccurs="0"/>` declarations inside `<xs:sequence>`. Two use unqualified `type="unsignedByte"`.	§§7.3 and 7.5 (pp. 74, 95) define `<distance>`, `<azimuth-direction>`, and `<elevation-direction>` as child elements. Declare them as `<xs:element ... minOccurs="0"/>` inside the sequence and qualify the numeric types as `xs:unsignedByte`.
P11	TS 24.545 V19.4.0	pp. 82, 91, §7.4.2	`srpos-info-req` uses `type="xs:SrPosInfoReqType"`; the local type is `tSrPosInfoReqType`.	Use `type="sealloc:tSrPosInfoReqType"`.
P12	TS 24.545 V19.4.0	p. 92, §7.4.2	Two `sealdatadelivery:anyExtType` references use an undeclared prefix.	Use the declared `sealloc:anyExtType`, if the locally defined extension type is intended.
P13	TS 24.545 V19.4.0	pp. 80, 90–91, 105, 107; §§7.3, 7.4.2, 7.5	Two location request types require a `requested-loc-type` attribute. §7.3 specifies a child element: mandatory for off-network configuration and optional for subscription. One type has no such child; the other has both child and required attribute.	Model the documented child element with the correct occurrence in each type and remove the conflicting required attributes.
P14	TS 24.545 V19.4.0	pp. 90, 105; §§7.4.2, 7.5	Schema child `history-loc-report-trigger` conflicts with the `<loc-report-trigger>` child and Table 7.5-4.	Name the schema child `loc-report-trigger`; remove the trailing space as part of that change.
P15	TS 24.545 V19.4.0	pp. 29–34, 83, 91, 93, 95; §§6.2.6, 7.4.2, 7.5	Two `expiry-time` declarations use `xs:nonPositiveInteger`, which excludes positive subscription lifetimes. §7.5 describes seconds; procedures reduce an accepted lifetime and explicitly use zero to delete a subscription.	Use `xs:nonNegativeInteger` in both declarations, preserving zero for the documented cancellation case.
P16	TS 24.545 V19.4.0	p. 90, §7.4.2, `tLocationReuseRequestType`	The `target-VAL-user-id` declaration begins `xs:element` without `<`; it becomes character content inside `xs:sequence` and fails schema compilation.	Restore `<xs:element .../>`. The opening bracket is absent in the rendered PDF, not lost during extraction.
P17	TS 24.545 V19.4.0	pp. 82–84, 89, §7.4.2	Global elements are declared repeatedly: `timestamp` three times; `velocity-info`, `VAL-user-id`, and `confirm` twice. The two velocity declarations also have different types.	Keep one global declaration per QName and reference it where needed. Reconcile the velocity type with its defined 14-hex-digit encoding; do not keep duplicate declarations even when their types agree.
P18	TS 24.545 V19.4.0	pp. 82, 84, 89, §7.4.2	Nine global element declarations carry `minOccurs` and, in four cases, `maxOccurs`; occurrence constraints are not allowed on global declarations.	Remove occurrence attributes from global declarations; express permitted multiplicity at local particles or in the extension semantics. Affects requested-velocity-info, velocity-info, VAL-user-id, slpos-mgmt-params, current-geographical-area, latest-geographical-area and timestamp declarations.
P19	TS 24.545 V19.4.0	pp. 82–83, 90–91, §7.4.2	Two `requested-pos-method` declarations reference `sealloc:tPositioningMethodType`, which is not defined in this schema. Related declarations use `xs:string`.	Define the intended positioning-method type or consistently use the existing string representation, with any allowed values explicitly specified.
P20	TS 24.281 V19.6.0, TS 24.282 V19.7.0, TS 24.379 V19.7.0, TS 24.545 V19.4.0	24.281/24.282 pp. 598, 600–601; 24.379 pp. 748, 750–751; 24.545 p. 87	XSD hexadecimal patterns contain literal `^` and `$`, e.g. `^[A-Fa-f0-9]{14}$`. XML Schema does not treat them as anchors: valid plain hexadecimal values are rejected.	Remove the anchor characters: use `[A-Fa-f0-9]{14}`, `[A-Fa-f0-9]{6}`, and `([A-Fa-f0-9]{4})` or `([A-Fa-f0-9]{6})` as the two alternatives. Keep the alternatives in one XSD pattern. Validator checks confirmed plain valid hex passes after the correction.

ID	Source	Location	Current text or conflict	Proposed correction
P21	TS 24.229 V19.7.0	pp. 517–518; §7.14.2.2, Table 7.14.1	The purported normative XML schema puts the instance element `<ss:ue-instance/>` directly inside `<xs:schema>`. This is not an allowed schema component and defines no UE identity element. Java rejects it as invalid/misplaced; the PDF rendering confirms the source.	Supply valid XML Schema declarations that implement §7.14.2.1 and reference or define the UE identity information from TS 24.174. Reconcile the extDEN target namespace and the simservs namespace explicitly. The exact replacement declaration needs standards author review; a top-level `xs:element ref` is not a valid mechanical fix.
P22	TS 24.481 V19.3.0	pp. 86–87; A.2.3, step13 NOTIFY example (caption incorrectly A.2.3-11; see N08)	XML uses `mgktp:` with no namespace declaration, never closes `<mgktp:GKTPs>`, and prints another `<d:element>` at the end instead of closing the existing element. The selector's `xmlns(...)` URI text does not declare an XML namespace.	Declare `xmlns:mgktp="urn:3gpp:ns:mcpttGKTP:1.0"` on an ancestor; add `</mgktp:GKTPs>`; replace the last `<d:element>` with `</d:element>`. After layout-only normalization the original fails with an unbound prefix; namespace repair exposes mismatched tags; all three repairs make the document parse.
P23	TS 24.282 V19.7.0	pp. 441, 631; §§19A.2.6, D.8.3	MBS semantics give literal status values `deleteing` and `not-deleteing`; the sending procedure uses `deleting` and parallel MCVideo/MCPTT definitions use the correctly spelled pair.	Use `deleting` and `not-deleting` consistently as the protocol values.
P24	TS 24.281 V19.6.0, TS 24.282 V19.7.0, TS 24.379 V19.7.0	24.281 pp. 440, 594; 24.282 p. 631; 24.379 p. 738	The MCVideo MBS de-announcement step selects the MBS type but then populates an MBMS body. Join semantics in all three specifications spell the MIME subtype `mms-usage-info+xml`.	Use the corresponding `mbs-usage-info+xml` MIME body consistently; coordinate the MCData service prefix with P01.
P25	TS 24.281 V19.6.0, TS 24.379 V19.7.0	24.281 pp. 162, 620, §§7.4.1.2.2, F.5.2; 24.379 p. 769, §F.5.2	Global declarations `multiplex-support`, `transmission-request`, and `mcptt-floor-request` carry `minOccurs`/`maxOccurs`. Java rejects both attributes on each global declaration.	Remove occurrence attributes from the global declarations. Preserve parent-context occurrence rules; if multiple request objects are intended, define a valid containing structure and its child cardinality.
P26	TS 24.281 V19.6.0, TS 24.282 V19.7.0, TS 24.379 V19.7.0	24.281 p. 279, §9.2.3.6.1.2; 24.282 p. 576, §25.6.1.2; 24.379 pp. 361, 681, §§10.1.3.6.1.2, 17.5.2	Four global `xs:element name="functional-alias"` declarations use `use="optional"`; Java rejects this attribute on an element declaration.	Remove `use="optional"`. Express optional occurrence in the containing particle or extension rules, not with `minOccurs` on a global declaration.
P27	TS 24.581 V19.2.0	p. 149, §9.2.3.1, Table 9.2.3.1-1	The Video SSRC of the Transmitting User field ID is printed as decimal `024` but binary `00010111` (23). IDs 000–022 are contiguous and 023 is otherwise unused.	Confirm one value and make both columns agree. The binary value 23 continues the sequence after Reception Mode (022); implementations that use 24 do not interoperate with those that use 23.
P28	TS 24.379 V19.7.0, TS 24.282 V19.7.0	24.379 pp. 202–203, 207; §§9.2.1.2 NOTE 2, 9.2.2.2.3; 24.282 pp. 158–159, 164; §§8.2.2 NOTE 1, 8.3.2.3	The client clause says a client no longer interested in any group needs no `application/pidf+xml` body: `Expires` zero "is sufficient". The participating function's procedure applies only to a PUBLISH that "contains an application/pidf+xml MIME body" (precondition 5), so a bodiless deaffiliation matches no server procedure.	Add the bodiless `Expires` zero PUBLISH to the preconditions of §9.2.2.2.3 (§8.3.2.3), with steps 9 to 13 skipped for it. OpenMCX accepts it and ends every affiliation of the client.
P29	TS 24.379 V19.7.0, TS 24.281 V19.6.0	24.379 pp. 624–625, §§16.2.2.4 step 3f, 16.2.2.5 step 3f; 24.281 pp. 506–507, §21.2.2.5 step 3f	The terminating participating MCPTT function sets the P-Asserted-Identity of a regroup creation notice to its own public service identity, but copies the incoming one into a removal notice. MCVideo sets its own identity on both. A client therefore cannot know which identity a removal notice carries.	Use the participating function's public service identity in §16.2.2.5 step 3f, as §16.2.2.4 and TS 24.281 §21.2.2.5 do. OpenMCX clients accept an MCPTT removal notice with either identity.
P30	TS 24.379 V19.7.0	p. 632, §16.2.4.1 step 7f	Step 7f uses "the list of MCPTT IDs for this terminating participating MCPTT function as generated in step 3)", but step 3 stores the regroup; the lists are made in steps 5 and 6.	Refer to steps 5) and 6).

Clause, figure, and table identifiers

ID	Source	Location	Current identifier or conflict	Proposed correction
N01	TS 24.379 V19.7.0	p. 195, §7A.6	Two distinct procedures are numbered `7A.6.2`; the first refers to deauthorisation in `7A.6.2`.	Number the second procedure `7A.6.3` in the body and contents; change the first procedure's forward reference to `7A.6.3`.
N02	TS 24.282 V19.7.0	pp. 155–156, §7A.3	Service-authorisation notification and migration deauthorisation are both numbered `7A.3.6`.	Number the second procedure `7A.3.7` in body and contents; preserve references to the authorisation notification at `7A.3.6`.
N03	TS 24.282 V19.7.0	p. 158, §7A.6	Two distinct procedures are numbered `7A.6.2`; the first refers to deauthorisation in `7A.6.2`.	Number the second procedure `7A.6.3` in the body and contents; change the forward reference to `7A.6.3`.
N04	TS 24.281 V19.6.0	p. 169, §7A.6.2	Deauthorisation step cites `7A.6.2`, although the deauthorisation procedure is already numbered `7A.6.3`.	Change the step's reference to `7A.6.3`.
N05	TS 24.282 V19.7.0	p. 539, §24.2.8	“Receiving notification of ad hoc group data communication redirection” is numbered `24.2.7.1`, already used under §24.2.7.	Number this subsection `24.2.8.1` in the body and contents.
N06	TS 24.545 V19.4.0	pp. 6, 62–65; §§6.2.24–6.2.25	CoAP subsections under two different parents are both labelled `6.2.26.3` and `6.2.26.4`, in the body and contents. The first `6.2.26.3` is titled “SLM client HTTP procedure” although its body is a CoAP procedure.	Number the first pair `6.2.24.3/.4` and the second pair `6.2.25.3/.4` in the body and contents; retitle the first as “SLM client CoAP procedure”. No cross-references use these numbers.
N07	TS 24.581 V19.2.0	pp. 8, 65, 94; §6.3.6.6	“Receive MCVideo call release - 2” is numbered `6.3.4.6.2`, already assigned on p. 65.	Number the p. 94 subsection `6.3.6.6.2` in the body and contents.
N08	TS 24.481 V19.3.0	p. 86, Annex A.2.3	Step 13 calls its SIP NOTIFY table `A.2.3-13`, but the caption repeats `Table A.2.3-11`.	Change the caption to `Table A.2.3-13`.
N09	TS 24.483 V19.2.0	p. 329, §10.2.97B3B10	Caption `Table 10.2.97B3B8.1` repeats the ID used two tables earlier, on p. 328.	Use `Table 10.2.97B3B10.1`.
N10	TS 24.483 V19.2.0	p. 401, §13.2.43A2	MCVideo caption `Table 5.2.48B84A2.1` is copied from the MCPTT table.	Use `Table 13.2.43A2.1`.
N11	TS 24.545 V19.4.0	pp. 120–121, §§B.2.3.19–20	`LocationReport` and `LocationInfo` both have caption `Table B.2.3.20-1`.	Change the `LocationReport` caption to `Table B.2.3.19-1`.
N12	TS 24.545 V19.4.0	p. 130, §B.3.1.2.4.3.1	FETCH payload caption repeats PUT table `B.3.1.2.3.3.1-2`.	Use `Table B.3.1.2.4.3.1-2`, as stated immediately above it.
N13	TS 24.545 V19.4.0	p. 137, §B.3.1.3.2	`LocationCapability` repeats the `UeInfo` heading/table ID `B.3.1.3.2.3`.	Use heading `B.3.1.3.2.4` and table `B.3.1.3.2.4-1`.
N14	TS 23.280 V19.10.1	p. 96, §10.1.5.6.1.1	Caption `Table 10.1.5.6.1-1` conflicts with prose and containing clause.	Use `Table 10.1.5.6.1.1-1`.
N15	TS 23.280 V19.10.1	p. 331, §10.16.6.2.1	Caption `Table 10.16.6.2-1` conflicts with nearby prose.	Use `Table 10.16.6.2.1-1`.
N16	TS 23.280 V19.10.1	p. 320, §10.15.3.4.2	Figure reference `10.15.3.4.x-1` retains a placeholder.	Use the captioned `Figure 10.15.3.4.2-1`.
N17	TS 23.280 V19.10.1	pp. 329–330, §10.16.5.3	Prose cites nonexistent `Figure 10.16.5.4-1`.	Use captioned `Figure 10.16.5.3-1`.
N18	TS 23.379 V19.11.0	pp. 34–35, §10.5.2.6	Prose cites `Table 10.5.2.1.6-1`; caption is `10.5.2.6-1`.	Cite `Table 10.5.2.6-1`.

ID	Source	Location	Current identifier or conflict	Proposed correction
N19	TS 23.379 V19.11.0	pp. 185–186, §10.9.1.3.5	Caption `Figure 10.9.1.3.5.1-1` adds a nonexistent subsection.	Match the prose: `Figure 10.9.1.3.5-1`.
N20	TS 23.379 V19.11.0	pp. 186–187, §10.9.1.3.6	Caption `Figure 10.9.1.6.1-1` differs from prose and section.	Use `Figure 10.9.1.3.6-1`.
N21	TS 23.379 V19.11.0	p. 258, §10.19.3.2.2	Release procedure cites setup in `10.19.3.2.x`.	Cite the preceding matching setup procedure, `10.19.3.2.1`.
N22	TS 24.229 V19.7.0	pp. 442, 502; §§7.2.23.7, 7.9A.9	`table 7.2.x-1` and `table 7.9A.y` retain placeholders.	Cite the matching tables immediately below: `Table 7.2.23-1` and `Table 7.9A.9-1`.
N23	TS 24.281 V19.6.0	pp. 226–227, 234–235, 260, 263	Five TNG2 expiry references and one emergency-call cancellation authorisation reference cite §6.3.3.1.16, which validates priority parameters.	Use §6.3.3.1.15 for the five timer references; use §6.3.3.1.12.4 for cancellation authorisation on p. 235.
N24	TS 24.282 V19.7.0	p. 441, §19A.3.2.2	Sending an initial MBS announcement twice cites nonexistent §19.3.2.2.	Cite the sending procedure §19.2.2.2 in both places.
N25	TS 24.379 V19.7.0	p. 386, §10.2.2.4.4.1	GROUP CALL ANNOUNCEMENT transport cites nonexistent §10.2.2.1.1.1.	Cite MONP transport §10.2.1.1.1, as comparable send steps do.
N26	TS 24.380 V19.2.0	pp. 176–193, §§8.2.4–8.2.17	Fourteen subtype descriptions cite nonexistent Table 8.2.2-1.	Cite subtype Table 8.2.2.1-1.
N27	TS 23.280 V19.10.1	pp. 344–345, 359; Table 10.17.2.10-1, §§10.17.2.13, 10.17.6.2	Two initial UE configuration citations say `Annex 6`; p359 also uses the wrong self-reference `[5]`, as does the p344 citation of Table A.6-1.	Use `Annex A.6 of the present document` for both Annex 6 citations and `table A.6-1 of the present document` on p344.
N28	TS 23.280 V19.10.1	p. 129, §10.3.2.2	Media distribution function is said to be defined in “subclause 7.4.2.3.5”, which does not exist in TS 23.280. 23.379 §7.4.2.3.5 (p. 25) is “Media distribution function”. Formerly D01.	Cite §7.4.2.3.5 of TS 23.379 `[16]`, the media distribution function definition.
N29	TS 24.379 V19.7.0	p. 324, §10.1.2.2.2.1	Priority-call authorisation cites nonexistent §6.2.8.8.1.8 twice. §6.2.8.1.8 (p. 76) is “Determining authorisation for originating a priority group call”, and the same wording elsewhere cites it. Formerly D02.	Cite §6.2.8.1.8 in both places.
N30	TS 24.545 V19.4.0	p. 68, §6.3.1.2.3	Timer expiry (N times) and acknowledgement/response reception both use `6.3.1.2.3.2` in the body.	Renumber the second body heading, acknowledgement/response reception, as `6.3.1.2.3.3`. The contents list only the parent clause.
N31	TS 33.180 V19.4.0	pp. 9, 140, Annex B	Token revocation and IdMS interface security are both labelled `B.12`.	Renumber IdMS interface security as `B.13` in the contents and body; keep token revocation as B.12.
N32	TS 23.280 V19.10.1	p. 340, §10.17.2.1; p. 367, §11.2.1	Prose cites `Table 10.17.2.1` although the caption is `Table 10.17.2.1-1`; signalling-control-plane text cites nonexistent `Figure 7.3.1-2`.	Cite `Table 10.17.2.1-1` and `Figure 7.3.1.2-2` respectively. The latter is captioned “Functional model for signalling control plane” on p39.
N33	TS 23.281 V19.10.0	p. 140, Figure 7.7.1.3.2C-1, step 4	Figure step 4 cites `subclause 7.7.1.3.2.B`; the matching textual step cites §7.7.1.3.2B, “End media reception – transmission control server initiated”.	Use `7.7.1.3.2B` in the figure.
N34	TS 23.379 V19.11.0	pp. 238–239, §10.19.2.17	Both prose (`Table 10.19.17 -1`) and caption (`Table 10.19.17-1`) omit the `.2` from the containing clause's number.	Use `Table 10.19.2.17-1` in both places; adjacent tables consistently follow §§10.19.2.16 and 10.19.2.18.
N35	TS 24.229 V19.7.0	p. 497; §7.9.9	The g.3gpp.rlos media feature-tag ASN.1 identifier remains `1.3.6.1.8.2.x`.	Use `1.3.6.1.8.2.37`, the value in the official IANA global-tree registry.

ID	Source	Location	Current identifier or conflict	Proposed correction
N36	TS 24.484 V19.6.0	pp. 99, 245; §§8.3.2.7, A.2.3	`<allow-to-receive-private-call-from-any-user>` cites table 8.3.2.7-28 (which is `<allow-regroup>`, p. 95) while the matching caption is 8.3.2.7-48; the HTTP GET step 11 points to A.23-11 while its caption is A.2.3-11.	Cite `Table 8.3.2.7-48` and `Table A.2.3-11`, respectively.
N37	TS 24.281 V19.6.0, TS 24.379 V19.7.0	24.281 p. 574; 24.379 p. 720; Annex E.2	The ICSI template numbers both Reference and Contact as `E.2.3`, then continues with E.2.4 Registration and E.2.5 Remarks.	Keep Reference as E.2.3; number Contact E.2.4, Registration E.2.5 and Remarks E.2.6; update contents accordingly.
N38	TS 24.281 V19.6.0, TS 24.379 V19.7.0	24.281 p. 286; 24.379 pp. 374, 378	Imminent-peril cancellation authorisation cites placeholder `6.2.8.1.A` in both specifications and `6.3.3.1.13.B` in MCPTT.	Use §6.2.8.1.26 in TS 24.281. In TS 24.379 use §§6.2.8.1.28 and 6.3.3.1.13.13, respectively, matching the client/server cancellation-authorisation titles.
N39	TS 24.282 V19.7.0	pp. 631–632, Annex D.8.3	MBS listening-status and eMBMS bearer semantics refer back to D.8.3. The listening-status sentence also names `<mbs-listening-status>` as its own inherited definition.	Refer to the MBMS definitions in D.5.3; the inherited listening element is `<mbms-listening-status>`. Keep `<mbs-listening-status>` as the new MBS element being defined, matching the parallel MCVideo/MCPTT clauses.
N40	TS 24.281 V19.6.0	pp. 220–266, 339–347, 373–377; authorisation definitions pp. 106–111	Eighty-three authorisation references cite nonexistent children §6.3.3.1.13.1–.6. §6.3.3.1.13 is a SIP 403 rejection procedure; the authorisation family is §6.3.3.1.12.1–.12.	Use the matching §6.3.3.1.12 procedure. Preserve the final number in 79 occurrences. For alert cancellation on pp. 374 and 376, change `.13.1` to `.12.3`; for imminent-peril initiation on pp. 234 and 258, change `.13.6` to `.12.5`.
N41	TS 24.281 V19.6.0	pp. 90, 220–267, 340–347; §§9.2.1.4, 9.2.2.4	A controlling-function block copied from TS 24.379 keeps the TS 24.379 numbering of §6.3.3.1, which TS 24.281 shifts down by one. SIP UPDATE for Resource-Priority correction is cited as §6.3.3.1.8 (TS 24.281: “Generating a SIP re-INVITE request”) in 17 places on pp. 259–267, and once as nonexistent §6.3.3.18 (p. 262); retrieving Resource-Priority values is cited as §6.3.3.1.19 (a SIP MESSAGE procedure) on pp. 90, 220, 259–264 and 340–347; the SIP 403 rejection is cited as §6.3.3.1.14 (imminent-peril re-INVITE) on pp. 224, 233, 258, 262, 341 and 343.	Cite §6.3.3.1.7 for the UPDATE steps, §6.3.3.1.18 for Resource-Priority values and §6.3.3.1.13 for the SIP 403 response. Keep §6.3.3.1.8 where a re-INVITE is generated (p. 105) and §6.3.3.1.19 for the emergency-alert receipt MESSAGE (pp. 392–395). Coordinate with N40.

Numbered bibliography references and copy text

ID	Source	Location	Current text	Proposed correction
R01	TS 23.280 V19.10.1	pp. 26, 52, 165, 260, 329, 387, 393	TS 23.237 `[9]`; TS 23.281 `[4]` (3); TS 23.282 `[5]` (3); MCVideo MapGroupToBearer attributed to TS 23.380 `[12]`; TS 33.180 `[19]`; four self-citations TS 23.280 `[16]`; TS 22.280 `[16]` (2) or `[2]` (4).	Use TS 23.237 `[10]`, TS 23.281 `[12]`, TS 23.282 `[13]`, TS 23.281 `[12]` for MCVideo MapGroupToBearer, TS 33.180 `[25]`, `the present document` for the four self-citations, and TS 22.280 `[3]`.
R02	TS 23.280 V19.10.1	pp. 21, 400; reference [34]	Entry [34] says `TS 25.501` but gives the 5G System architecture title, while Annex E cites `TS 23.501 [34]`.	Correct entry [34] to `TS 23.501`.
R03	TS 23.281 V19.10.0	pp. 48, 56–57, 80, 168, 203, 218, 220, 223–225, 227, 229	TS 23.280 `[16]`/`[5]`/`[18]` (10); TS 22.280 `[16]`/`[17]` (13); TS 33.180 `[19]`/`[18]` (2), plus one TS 33.180 citation without an index; §7.19.3.2.1 cites nonexistent §7.17.3.2.4.	Use TS 23.280 `[6]`, TS 22.280 `[2]`, TS 33.180 `[14]` (including the missing index), and participant-update §7.19.3.2.2.
R04	TS 23.282 V19.10.0	pp. 88, 137, 242, 245, 284, 286, 291, 293, 295, 298	TS 23.280 `[16]`/`[6]` (8); TS 22.280 `[17]` (7); TS 33.180 `[19]`; TS 22.282 `[2]`/`[5]` (2); nonexistent §§7.17.3.15 and 10.17.3.1.4.	Use TS 23.280 `[5]`, TS 22.280 `[2]`, TS 33.180 `[13]`, TS 22.282 `[3]`, and §§7.17.3.1.5 and 7.17.3.1.4 respectively.
R05	TS 23.379 V19.11.0	pp. 94, 110, 142, 227, 275	TS 23.280 cited as `[5]`/`[8]`; TS 33.180 cited as `[11]` in Annex A although [11] is Void.	Use `[16]` for TS 23.280 and `[19]` for TS 33.180.

ID	Source	Location	Current text	Proposed correction
R06	TS 24.229 V19.7.0	pp. 63, 66, 145, 190, 197, 245, 341, 458, 498, 527, 1041–1043	TS 22.101 [8]; TR 33.203 [19]; RFC 4412 [16]; RFC 3261 [6]; RFC 7459 [225]; TS 33.203 [9]; RFC 979 [302]; twice RFC 3329 [4]; RFC 3840 [53] with wrong security subclause 12.1; RFC 5917 [192]; six TS 24.501 [257] citations.	Use TS 22.101 [1A], TS 33.203 [19], RFC 4412 [116], RFC 3261 [26], RFC 7549 [225], TS 33.203 [19], RFC 9795 [302], RFC 3329 [48], RFC 3840 [62] §11.1, RFC 6917 [192], and TS 24.501 [258], respectively.
R07	TS 24.281 V19.6.0	pp. 167, 343, 439–443, 489, 533, 546–549, 560–561	TS 24.229 [4] (18 instances), RFC 3428 [33], and RFC 3841 [6] have unrelated bibliography entries. The citations occur in MBS steps and also in key download, authorisation notify, and other procedures.	Use TS 24.229 [11], RFC 3428 [17], and RFC 3841 [20] wherever they occur.
R08	TS 24.282 V19.7.0	pp. 156, 440–443, 491, 539, 557, 560	TS 24.229 [4] (15 instances) or [11], RFC 3428 [33] or [17], and RFC 3841 [6]/[20] have unrelated bibliography entries; [17] is RFC 4975. The citations occur in MBS steps and in other procedures.	Use the local entries for TS 24.229 [5], RFC 3428 [6], and RFC 3841 [8] wherever they occur.
R09	TS 24.482 V19.1.0	p. 7, reference [22]	Entry labels TS 23.282 “MCData signalling control,” but the body cites TS 24.282 [22].	Identify reference [22] as TS 24.282.
R10	TS 24.484 V19.6.0	pp. 149, 210, 212	TS 24.4283 [4]; MCData table row cites MCVideo TS 24.281 [25]; TS 24.483 [3].	Use TS 24.483 [4], TS 24.282 [25], and TS 24.483 [4] respectively.
R11	TS 24.545 V19.4.0	pp. 20–56, 94–104, 122–123, 149	Nine CoAP authorisation steps cite HTTP boot-up §6.2.1.2; TS 23.032 cited as [2]; LocationQoS cited as TS 29.546 [29]; general error responses cited as TS 24.546 [23], while [23] is RFC 7641.	Cite CoAP identity §6.2.1.3, TS 23.032 [3], and TS 24.546 [29] for both wrongly numbered TS 29.546 citations and the wrongly indexed TS 24.546 citation.
R12	TS 24.545 V19.4.0	pp. 53, 64, 157	CoAP RFC 7252 [14] and CBOR RFC 8949 [20] use indexes assigned to other RFCs.	Use RFC 7252 [21] and RFC 8949 [26].
R13	TS 24.581 V19.2.0	pp. 18, 58, 107, 230–231	TS 24.481 [12] (three instances), TS 33.180 [18], RFC 3711 [16] (two), and TS 24.281 [12] refer to unrelated or absent bibliography entries.	Use TS 24.481 [5], TS 33.180 [8], RFC 3711 [4], and TS 24.281 [2] respectively.
R14	TS 33.180 V19.4.0	pp. 23, 59, 70, 122, 142, 169, 173–174, 181	TS 23.228 [23], RFC 6509 [8], TS 23.281 [39], TS 33.203 [9], TS 24.379 [10], TS 33.220 [27], and Annex A.1 TS 23.179 [2] have mismatched numbers or names.	Use TS 23.228 [15], RFC 6509 [11], TS 23.281 [37], TS 33.203 [6], TS 24.379 [49], TS 33.220 [17], and TS 23.379 [2] respectively. Annex A.11–A.12 remain separate in D06.
R15	TS 24.281 V19.6.0	pp. 487–488	Three citations of TS 24.229 [5]; [5] is TS 24.581.	Use TS 24.229 [11].
D07	TS 23.280 V19.10.1, TS 23.379 V19.11.0	23.280 p. 114, §10.2.4.1; 23.379 p. 100, §10.6.2.11.1	Temporary broadcast regroup requirements are cited as TS 22.180 [2] and TS 22.280 [2]; both documents' [2] identifies TS 22.179, whose relevant requirements are now Void.	Cite TS 22.280 [3] in TS 23.280 and TS 22.280 [17] in TS 23.379. Its Release 19 §6.6.3 contains the active temporary broadcast group requirements.
R16	TS 26.281 V19.0.0	pp. 6, 8, §4.1.3	User-plane text assigns RTP to [7], [9] and SRTP to [8], [10]; the bibliography identifies [8] as RTP and [9] as SRTP. [7] is TS 33.179 and [10] is the H.264 RTP payload format.	Cite RTP as [8] and SRTP as [9]. Cite H.264 RTP payload format [10] and security [7] separately where those subjects are intended. This replaces the earlier statement that no candidate was found in TS 26.281.
R17	TS 33.180 V19.4.0	pp. 23, 181, 185; §4.3.4.2, Annex I, Table J.1.3-1	RFC 5853 [24] occurs three times, but [24] is RFC 4771. RFC 4122 [14] describes UUIDs, but [14] is TS 33.401. Neither intended RFC has a bibliography entry.	Add bibliography entries for RFC 5853 and RFC 4122 and cite their assigned indexes; do not reuse an unrelated existing number.

ID	Source	Location	Current text	Proposed correction
R18	TS 24.229 V19.7.0	p. 1003; Q.2.2.1	Two mentions of RFC 3315 [40] in DHCPv6/FAP configuration, but [40] p. 49 is RFC 8415. RFC 8415 obsoletes RFC 3315. These occurrences do not cite specific RFC sections.	Report the stale-name/bibliography mismatch. Align the text with the chosen DHCPv6 normative reference (likely RFC 8415 [40]); if the older document is deliberately needed, cite it separately. Do not characterize it as demonstrated wrong DHCPv6 behavior.
R19	TS 24.281 V19.6.0	pp. 41, 164, 172, 175, 198, 286, 487, 489, 526, 531, 538–539, 541, 546, 560, 582, 594–595	Additional mismatched citations: RFC 8101 [48]; TS 24.484 [50]; RFC 3903 [37]; RFC 6665 [26]/[22]; RFC 6050 [9]/[7]; RFC 3856 [51]; RFC 3261 [24]; TS 24.380 [5]; TS 29.468 [42].	Use RFC 8101 `[38]`; TS 24.484 `[25]`; RFC 3903 `[12]`; RFC 6665 `[16]`; RFC 6050 `[14]`; RFC 3856 `[13]`; RFC 3261 `[15]`; TS 24.581 `[5]`; and TS 29.468 `[44]`, respectively.
R20	TS 24.282 V19.7.0	pp. 37, 97–98, 160, 352, 491, 533, 557, 573–576, 591, 632, 643	Additional mismatched citations: RFC 4575 [KK]; RFC 3903 [37]; TS 24.483 [4]/[2]; RFC 6050 [9]; TS 24.484 [50]; RFC 3856 [51]; RFC 4538 [32]; RFC 3261 [24]; RFC 6665 [52]; TS 24.481 [31]; RFC 4483 [32]; TS 29.468 [42].	Use RFC 4575 `[87]`; RFC 3903 `[34]`; TS 24.483 `[42]`; RFC 6050 `[7]`; TS 24.484 `[12]`; RFC 3856 `[39]`; RFC 4538 `[54]`; RFC 3261 `[4]`; RFC 6665 `[36]`; TS 24.481 `[11]`; RFC 4483 `[13]`; and TS 29.468 `[57]`, respectively.

Wording, diagrams, and editorial normalization

ID	Source	Location	Current text	Proposed correction
E01	TS 24.484 V19.6.0	pp. 108, 220, 223	Schema prints `name="default-pqi "`; prose twice prints `<default-pqi >`.	Remove the stray spaces. This is source-text hygiene: a Java XML Schema validator accepted the schema declaration and `<default-pqi/>` after whitespace normalization.
E02	TS 24.281 V19.6.0, TS 24.282 V19.7.0, TS 24.379 V19.7.0	24.281 pp. 599, 620; 24.282 p. 599; 24.379 p. 748	Schema names have leading spaces before `RRC_INACTIVE_PersistencePeriod` and, in 24.281, `transmission-request-Type`. After removing the separate illegal root occurrence attributes, Java could not resolve the unspaced reference to the latter type; trimming its declaration made the schema compile.	Trim the names. XML Schema NCName whitespace normalization prevents treating every space as a distinct wire name; the named-type case has a reproduced Java compatibility effect.
E03	TS 24.281 V19.6.0, TS 24.282 V19.7.0, TS 24.379 V19.7.0	24.281 p. 169; 24.282 p. 158; 24.379 p. 195	`primary-mcvideo-id>`, `primary-mcdata-id>`, and `primary-mcptt-id>` are printed without opening `<`.	Restore `<primary-mcvideo-id>`, `<primary-mcdata-id>`, and `<primary-mcptt-id>`.
E04	TS 23.281 V19.10.0, TS 23.282 V19.10.0	23.281 pp. 179, 186; 23.282 pp. 237, 258	MCVideo tables describe the actor as “MCPTT server”. A MCData figure labels “Home MCPTT provider”; another has four MCVideo-client labels in its steps.	Use “MCVideo server” in the two MCVideo tables, “Home MCData provider” on MCData p. 237, and “MCData client” in all four affected labels on MCData p. 258.
E05	TS 23.280 V19.10.1, TS 23.281 V19.10.0	23.280 p. 374; 23.281 p. 57	Figure 11.5.2.3.3-1 is titled “On-demand location reporting” under the cancel procedure; a MCVideo regroup figure points to TS 23.380 while its step text points to TS 23.280.	Title the first “Location reporting cancel procedure”; cite TS 23.280 in the regroup figure.
E06	TS 24.545 V19.4.0	pp. 15, 63–66	SLM-S requirements say SLM-C. Subscription response/request roles are swapped; §6.2.25.2 addresses its request to SLM-S although it sends to SLM-C. On p. 66, the SLM-S response is sent to itself and SLM-S is named as generator of the client POST.	Align each actor with its procedure role. The p. 64 subscription Request-URI must identify the target SLM-C; the p. 66 HTTP response recipient and notification POST generator are SLM-C.
E07	TS 24.482 V19.1.0	p. 17, Annex A.2.2.3	Trusted-entity TCP connections are said to arrive at the FQDN/port “for UEs” after configuring the trusted-entity listener.	Name the trusted-entity FQDN/port.
E08	TS 24.484 V19.6.0	pp. 42, 97, 115, 186	Prose tags contain a space before `>`: `<SNSSAI-Info >`, `<anyExt >`, `<allow-cancel-private-call-call-back >`, `<max-duration-without-floor-control >`, and `<LocationCriteriaForDeactivation >`.	Remove spaces for consistent element notation. Whitespace before `>` is legal XML, so this is editorial. The similar p. 267 occurrence is a historical CR title.

ID	Source	Location	Current text	Proposed correction
E09	TS 24.282 V19.7.0, TS 24.379 V19.7.0, TS 24.545 V19.4.0	24.282 pp. 538, 626; 24.379 p. 685; 24.545 pp. 66, 93	Prose tags contain spaces before `>`: `<msgstore-ctrl-command-list >`, `<mcdata-calling-group-id >`, `<mcptt-calling-group-id >`, `<location-positioning-notification >`, and `<latest-location >`.	Normalize displayed tag notation. The hyphen in `command-list` is present across the PDF line break; no separate wire value is implied.
E10	TS 23.280 V19.10.1	p. 96, §10.1.5.6.1.2	Subscribe-group-dynamic-data-response prose ends the third destination with `and from the MC service server to the.`	Complete the destination as `location management server`, matching the request's origin in §10.1.5.6.1.1 and the actual response flow in §10.1.5.6.4.1 on pp100–101.
E11	TS 23.280 V19.10.1	pp. 100–101, §§10.1.5.6.4.1–.2; contents p. 9	Subsection titles run into body text: `SubscriptionThe procedure` and `NotificationThe procedure`; the contents omit these entries, as they omit the sibling entries 10.1.5.6.2.1–.2 and 10.1.5.6.3.1–.2.	Restore separate numbered headings `10.1.5.6.4.1 Subscription` and `10.1.5.6.4.2 Notification`, with body paragraphs below, and include the subsections in the contents consistently. The PDF rendering confirms this is source formatting rather than extraction alone.
E12	TS 23.379 V19.11.0	p. 129, §10.7.2.4.1, NOTE 5	Emergency alert cancellation refers to `3GPP TS 23.280 subclause 10.10.1.2.2.2` without a bibliography index.	Add the local reference `[16]` after `TS 23.280`. This is a citation-format omission; the document number and target clause already identify the destination.

Gaps needing a standards decision

ID	Source	Location	Gap	Decision needed
D03	TS 24.545 V19.4.0	pp. 63–65, 144; §§6.2.24–6.2.25, B.4.1	`6.2.q.4` and two `B.4.1.2.Y` references remain placeholders. B.4.1 explicitly says subscription API updates are FFS. Request/response object names and MIME strings differ (including `seal-data-delivery-info` and `subsription`); the success response is both 2.01 and 2.05.	Resolve `6.2.q.4` with N06 (likely §6.2.24.4), then complete and specify the subscription resource, object definitions, media types and response status consistently. Do not silently substitute the configuration resource for the undefined subscription API.
D04	TS 24.281 V19.6.0, TS 24.282 V19.7.0	24.281 pp. 414–415, 417, 419, 421, 526; 24.282 pp. 533, 548–550	Editor's notes leave ambient-viewing/pre-established-session procedures, terminating procedures, and FD parameters for future definition.	Complete the procedures/parameters or delimit the unsupported feature scope.
D05	TS 24.582 V19.0.0	p. 14, §6.1.1.2.1	Originating passive MSRP endpoint sends no initial empty SEND, while payload sending waits for a 200 response to that SEND.	Confirm the passive-originator trigger and define it if missing. This is a protocol lead, not a proven deadlock.
D06	TS 33.180 V19.4.0	p. 124, Annexes A.11–A.12	Three TS 23.179 `[2]` citations name a document absent from the bibliography; cited §8.2 does not exist in TS 23.379 `[2]`.	Check each intended target. TS 23.280 `[36]` contains relevant identity (§8.2) and deployment (§9) material.
D08	TS 24.545 V19.4.0	pp. 29, 73, 83, 87, 93; §§6.2.6, 7.3–7.5	`req-time-info` is defined as time information, but its XSD type is `tVelocityInfoType`, a 14-hex-digit velocity encoding.	Specify the intended time representation and assign the matching schema type. A timestamp, duration or interval cannot be selected from the existing text with sufficient confidence.
D09	TS 24.481 V19.3.0	p. 70; §7.4.2 (group key transport payload structure)	Editor's note says XML element(s)/attribute(s) for the KMS URI in the MCPTT Group “need to be defined”, while the preceding NOTE says the KMS URI is stored in the group configuration data.	Resolve the explicit configuration gap or replace the note with a concrete existing representation after checking the intended group document schema. No invented field name proposed.
D10	TS 24.484 V19.6.0	pp. 22, 195, 213; §§4.2.3.1, 10.3.2.7, 10.4.2.1	Three explicit editor's notes: identities element is FFS; MCData `<kms-sec>` / `KMSSEC` configuration needs specification; signalling/media protection mechanisms are yet to be agreed and configuration expected to be added.	Author review is needed to decide whether these are still real omissions or stale editor's notes. They should be recorded as explicit unresolved notes, not asserted as proven missing functionality.

ID	Source	Location	Gap	Decision needed
D11	TS 24.281 V19.6.0	pp. 326–327, 343–345, 375–376	Seven MCVideo steps (two client, five server) say to interact with the media or user plane under TS 24.380 [79], whose §6.3 specifies MCPTT floor control. The index matches the bibliography, but parallel MCVideo steps cite TS 24.581 [5] for transmission control.	Confirm whether these seven steps should cite TS 24.581 [5]. Do not replace all references to TS 24.380: the transmission-priority encoding reference in F.5.3 may be intentional.